

SIMATS ENGINEERING

ASSESSMENT TOOL 3

Name	Abinеш M
Register Number	192465050
Date	
Time	1 Hour

COURSE NAME: CYBERSECURITY INCIDENT HANDLING AND RESPONSE
COURSE CODE: CSA6001

COURSE OUTCOME COVERED

CO1: Apply the concepts of cybersecurity incidents by distinguishing security events from incidents, implementing incident handling lifecycle stages, frameworks, with documentation practices in organizational contexts. (BL1, BL2)

Activity Tool : Technical Presentation (Low)
Assessment Tool Weightage: 10%

Code of Conduct:

*I Abinеш M - 192465050 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weight age	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Evidence Acquisition and Integrity Verification	25%	Correctly acquires evidence, calculates SHA-256 hash before and after analysis, and verifies integrity without errors.	Correctly performs evidence acquisition and hash verification with minor procedural errors.	Performs evidence acquisition with incomplete or partially correct integrity verification.	Fails to acquire evidence properly or verify integrity using hash values.
Digital Evidence Analysis	25%	Accurately identifies all relevant digital artifacts (e.g., logs, browser history, suspicious files) and provides correct forensic interpretation.	Identifies most relevant artifacts with minor omissions or interpretation errors.	Identifies only some relevant artifacts with limited analysis.	Fails to identify relevant digital evidence or provides incorrect analysis.
Chain of Custody Documentation	20%	Completes the Chain of Custody form accurately with all required details (Case ID, date/time, collector, description, hash value, transfer record, signature).	Completes most sections correctly with minor omissions or formatting errors.	Documentation is partially complete with several missing or inaccurate details.	Chain of Custody documentation is incomplete, incorrect, or missing.
Application of Forensic Tools and Procedures	20%	Effectively uses Autopsy/Sleuth Kit and Linux commands (sha256sum, cp, ls -l) following standard forensic procedures.	Uses forensic tools correctly with minor procedural mistakes.	Demonstrates basic use of tools but requires guidance or shows procedural gaps.	Unable to use forensic tools correctly or follow forensic procedures.
Presentation and Technical Reporting	10%	Findings are clearly documented, technically accurate, well-organized, and professionally presented.	Report is generally clear and organized with minor technical or language issues.	Report is understandable but lacks organization or technical clarity.	Report is incomplete, poorly organized, or difficult to understand.

Criteria	Evidence Acquisition and Integrity Verification (4)	Digital Evidence Analysis (4)	Chain of Custody Documentation (4)	Application of Forensic Tools and Procedures (4)	Presentation and Technical Reporting (4)	TOTAL (20)
Weightage	30%	25%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

3. Unauthorized USB Usage Investigation

Aim

To perform a digital forensic examination of a USB storage device suspected of being used to copy confidential project documents by an employee before resignation, using Autopsy or The Sleuth Kit (TSK). The investigation includes identifying copied files, analyzing timestamps, verifying evidence integrity using SHA-256 hashing, and documenting the handling of evidence through a Chain of Custody form.

Objective

- Acquire and analyze the forensic image of the USB device.
- Identify confidential files copied onto the USB drive.
- Examine file metadata and timestamps.
- Verify evidence integrity using SHA-256 hash values.
- Maintain proper evidence handling documentation.

Tools Required

- Autopsy Digital Forensics Tool
- The Sleuth Kit (TSK)
- FTK Imager (for acquiring images)
- SHA-256 Hash Calculator (sha256sum, HashCalc, or CertUtil)
- Write Blocker (Hardware/Software)
- Windows/Linux Operating System

Scenario

A software engineer submits a resignation. The security team notices unusual USB activity during the employee's final working day. System audit logs indicate that several confidential project files were accessed shortly before the employee connected a personal USB device.

The USB device is seized and a forensic image is created. Investigators must determine whether confidential files were copied, identify the copied documents, verify their integrity, and document the investigation.

Forensic Investigation Process

Step 1: Evidence Acquisition

The first step is to create an exact forensic image of the USB storage device.

Procedure

1. Connect the USB through a write blocker.
2. Launch FTK Imager.
3. Select Create Disk Image.
4. Choose Physical Drive.
5. Select the USB device.
6. Save the image as:
 - RAW (.img)
 - E01 format
7. Calculate SHA-256 hash during acquisition.

Purpose

- Preserve original evidence.

- Prevent accidental modification.
- Enable repeatable forensic analysis.

Step 2: Evidence Preservation

After acquisition:

- Store original USB securely.
- Analyze only the forensic image.
- Record acquisition details.
- Verify image integrity.

Step 3: Create Case in Autopsy

1. Open Autopsy.
2. Click Create New Case.
3. Enter:
 - Case Name
 - Investigator Name
 - Description
4. Click Next.
5. Add Data Source.
6. Select Disk Image.
7. Choose USB image.

Autopsy begins indexing the evidence.

Step 4: File System Examination

Autopsy displays:

- File System
- Deleted Files
- Recent Files
- Documents
- Images
- Videos
- Metadata

Navigate to:

Data Sources



USB Image



Documents

Look for:

- DOCX
- PDF
- XLSX
- PPTX
- TXT

Step 5: Identify Confidential Documents

Example findings:

File Name	File Type	Size
ProjectPlan.docx	DOCX	3.5 MB
ClientData.xlsx	XLSX	5 MB
FinancialBudget.pdf	PDF	2 MB
SourceCodeBackup.zip	ZIP	25 MB

These files may contain confidential company information.

Step 6: Metadata Analysis

Every file contains metadata.

Example metadata includes:

- File name
- Owner
- File path
- Created time
- Modified time
- Last accessed time
- File size
- Extension

Example:

File Name:

ProjectPlan.docx

Location:

Documents/Projects/

Created:

2026-07-15 10:05

Modified:

2026-07-25 16:20

Accessed:

2026-07-25 16:25

Size:

3.5 MB

Step 7: Timestamp Analysis

Timestamp analysis determines when activities occurred.

Three important timestamps:

Created Time

Indicates when the file was originally created.

Example:

Created:

15 July 2026

Modified Time

Indicates when the file content was last changed.

Example:

Modified:

25 July 2026

Accessed Time

Shows when the file was last opened.

Example:

Accessed:

25 July 2026

Investigators compare these timestamps with:

- Employee login records
- USB insertion logs
- Office attendance
- Resignation date

If timestamps match suspicious activity, evidence becomes stronger.

Step 8: USB Connection Logs

System logs may indicate:

USB Connected

Time:

4:18 PM

User:

Employee01

Device ID:

Kingston USB

Later:

ProjectPlan.docx accessed

4:20 PM

Copied to USB

Timeline:

4:18 PM

USB inserted

↓

4:20 PM

ProjectPlan.docx opened

↓

4:21 PM

ClientData.xlsx copied

↓

4:25 PM

USB removed

This sequence indicates potential data exfiltration.

Step 9: SHA-256 Hash Verification

Hashing verifies that files remain unchanged during investigation.

Example command (Linux):

sha256sum ProjectPlan.docx

Example output:

d1b7af61b65a5ed6f12ef4e39df6f74a56e2a31c6e56e0d2d471d8b6d9c4d251

Windows:

certutil -hashfile ProjectPlan.docx SHA256

Compare:

Original Evidence

Same Hash Integrity Verified

If hashes differ:

- Evidence modified
- Investigation compromised

Step 10: Reporting Findings

Example Report

Case ID: USB-2026-001

Evidence:

- Kingston 32GB USB

Recovered Files:

- ProjectPlan.docx
- ClientData.xlsx
- FinancialBudget.pdf

Timeline:

- USB inserted
- Confidential files opened
- Files copied
- USB removed

Hash Verification:

SHA-256 values matched.

Conclusion:

Evidence indicates confidential company documents were copied onto the USB device before resignation.

Chain of Custody

Purpose

Maintains accountability for evidence from seizure to court presentation.

Sample Chain of Custody Form

Field	Details
Case ID	USB-2026-001
Evidence ID	E-101
Description	Kingston 32GB USB Drive
Serial Number	KNG123456

Field	Details
Collected By	Investigator A
Date	25-07-2026
Time	05:30 PM
Location	Employee Workstation

SHA-256 Hash d1b7af61...

Transferred To Digital Forensic Lab

Purpose Evidence Analysis

Received By Examiner B

Storage Evidence Locker

Observations

- Confidential files recovered.
- Metadata confirmed recent access.
- Timeline matched USB insertion.
- SHA-256 hash verified evidence integrity.
- Chain of Custody maintained throughout.

Result

The forensic examination successfully identified confidential documents copied onto the USB storage device. File metadata and timestamps confirmed that the files were accessed and copied shortly before the employee's resignation. SHA-256 hash verification confirmed that the evidence remained unaltered during the investigation. Proper Chain of Custody documentation ensured the admissibility and integrity of the digital evidence.

4. Malware Investigation.

Aim

To investigate a malware-infected workstation using Autopsy or The Sleuth Kit, identify the suspicious executable responsible for the infection, calculate and verify SHA-256 hash values, and document evidence handling using the Chain of Custody.

Objective

- Analyze the forensic disk image.
- Locate suspicious executable files.
- Examine file metadata and execution timeline.
- Verify evidence integrity using SHA-256.
- Maintain Chain of Custody documentation.

Scenario

A workstation is isolated after antivirus software reports unusual activity. Network logs show communication with suspicious IP addresses. A forensic disk image is acquired, and investigators suspect a malicious executable is responsible for the infection.

Investigation Process

Step 1: Acquire Disk Image

- Create a forensic image using FTK Imager or dd.
- Record SHA-256 hash.
- Store the original disk securely.

Step 2: Load Image into Autopsy

- Create a new case.
- Add the forensic disk image.
- Allow indexing to complete.

Step 3: Identify Suspicious Executables

Search for executable files (.exe, .dll, .bat, .ps1) in:

- Startup folders
- Temp directories
- Downloads
- AppData

Example suspicious file:

svchost_update.exe

Location:

C:\Users\User\AppData\Roaming\

Step 4: Analyze Metadata

Review:

- File size
- Creation date
- Modification date
- Last accessed date

- Digital signature
- File path

Unsigned executables in unusual locations are highly suspicious.

Step 5: Timeline Analysis

Autopsy timeline may show:

09:30 AM – Malicious executable created

09:31 AM – Executed

09:32 AM – Registry modified

09:34 AM – Suspicious network connection established

This sequence helps establish the infection timeline.

Step 6: SHA-256 Hash Verification

Linux:

```
sha256sum svchost_update.exe
```

Windows:

```
certutil -hashfile svchost_update.exe SHA256
```

Compare hashes before and after analysis. Matching values confirm the executable was not altered.

Step 7: Document Findings

Example:

- Suspicious file: svchost_update.exe
- Location: AppData\Roaming
- First execution: 09:31 AM
- Registry persistence detected.
- SHA-256 verified.

Chain of Custody

Field	Details
Case ID	MAL-2026-001
Evidence ID	E-202
Evidence	Forensic Disk Image
Collected By	Investigator A
Date	26-07-2026
Time	10:00 AM
SHA-256 Hash	9fa34d...
Transferred To	Malware Analyst
Purpose	Malware Investigation
Storage	Digital Evidence Vault

Observations

- Suspicious executable identified.
- File metadata analyzed.

- Timeline correlated with malware activity.
- SHA-256 hashes matched before and after analysis.
- Evidence integrity preserved through proper handling.

Result

The forensic analysis successfully identified the suspicious executable responsible for the malware infection. Timeline analysis, metadata examination, and hash verification confirmed the executable's role in the incident while demonstrating that the evidence remained unchanged. The Chain of Custody was maintained throughout the investigation, ensuring the evidence's integrity and legal admissibility.